

KRITHIKA NITHYANANDAM

Software Engineer · Generative AI · Platform Engineering · M.Tech AI & ML (BITS Pilani)

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SUMMARY

Software engineer specializing in **Generative AI systems and platform engineering**. Experience building LLM-powered applications, agent frameworks, and developer tooling with real-time interfaces and production observability. Technical writer and speaker on AI topics. Currently pursuing an M.Tech in Artificial Intelligence and Machine Learning at BITS Pilani.

EXPERIENCE

Engineer – Generative AI & Platform | Presidio

Aug 2024 – Present

Architecting agentic frameworks and intelligent application platforms.

SpecifAI (AI-powered requirement generation platform)

- Eliminated a Python AI backend by consolidating LLM integrations into the **Electron application**, completing the migration in four days with zero defects and unblocking a critical product release.
- Integrated OpenRouter and Ollama to enable **multi-model orchestration** across LLM providers.
- Built user-scoped **BYOK LLM credential flows** for per-user model access.
- Built an **AI-driven requirement generation workflow** with conversational retrieval across requirements.
- Engineered **inline editing directives** enabling requirements to be created and updated directly through chat.
- Integrated **LangFuse and PostHog** for token usage tracking, latency monitoring, and observability.

HAI Framework (agent framework and reusable agent library)

- Built the **HAI Playground**, enabling agent instances to be wrapped and automatically exposed through a WebSocket-driven runtime and interactive UI for real-time experimentation.
- Developed **pluggable conversation history adapters** supporting database and filesystem storage.
- Built an AWS Marketplace product tour enabling guided onboarding for enterprise users.

AI Studio (LLM workspace platform with chat and task-driven agents)

- Primary developer of the **Superadmin module**, including the LiteLLM cost governance dashboard.
- Built the **Admin Portal** for agent and connector management with managed authentication.
- Built the **Assistant Stepper UI** and template-led connector setup, enabling structured agent creation while preserving flexible, tool-level configuration for MCP-based integrations.
- Delivered **agent templates, workspace creation, initial chat settings, quick actions, browser notifications, and connectors UI** as end-to-end platform capabilities.
- Built a **File Manager** and **File Suite** supporting CSV and Excel previews (PapaParse, ExcelJS) and PPT with multi-file workflows; completed a long-standing library migration backlog item for file-tree operations.
- Authored **DESIGN.md** to establish LLM-optimized UI design rules and agent-generation standards across the platform.
- Owned **AI Studio documentation and onboarding**, including Terraform-managed docs infrastructure (AWS S3, CloudFront, Route53) and a custom in-product tour (migrated from Driver.js to bespoke implementation).
- Contributed to **WebSocket to API migration** and resolved a long-standing **EFS conflict** between develop and ephemeral environments.

HAI Build (agentic code generation platform)

- Built the **SpecifAI MCP Server** with test case tools and project-path resolution.
- Resolved critical terminal crashes by implementing a **PTY truncation fix** for VS Code limits.
- Implemented **Fuse.js powered tree views** with real-time project reindexing.

Associate Engineer – DraftKings Mobile | Presidio

Oct 2023 – Aug 2024

- Delivered **Braze SDK integration** two days ahead of schedule, receiving client commendation.
- Resolved **64 production issues**, including six P1 incidents, within a high-pressure release environment.
- Built a **Network Performance Dashboard** using New Relic telemetry to monitor mobile application performance and diagnose production issues.
- Authored **Startup Manager documentation** and delivered demos for 70+ stakeholders.

WRITING

- **The Ephemeral Ledger: Keeping AI Agents Aligned Until the Feature Ships** – Proposed a lightweight, branch-local state ledger for keeping AI coding agents aligned across tools and sessions. *Presidio HAI, Medium.*
- **The Brownfield Gap: When AI Meets an Existing Codebase** – On context starvation and the gap between plausible code and your code. *Presidio HAI.*
- **The Psychology of LLMs: The Hidden Engineering Behind Empathy, Creativity and Identity** – Exploring how people understand, trust, and interact with language-model systems. *Presidio HAI.*
- **VS Code Extension Storage Explained: The What, Where, and How** – Deep dive into global and workspace-scoped storage mechanisms for VS Code extensions. *Medium.*
- **The Elegant Architecture of PTYs – Behind Your Terminal** – On pseudo-terminals in Unix-like systems and the software layer that powers interactive terminal sessions. *Medium.*

SPEAKING & MENTORSHIP

- **MCP Masterclass, Presidio** – Delivered an internal masterclass on Model Context Protocol, covering architecture, tool design, and practical integration patterns.
- **QA Enablement Sessions, Presidio** (Chennai, Coimbatore & Bengaluru) – Delivered applied AI enablement covering Model Context Protocol and Prompt Caching for QA audiences.
- **Pre-placement Talk, LICET** – Shared career and industry perspectives with students preparing to enter software engineering roles.
- **PRIME Internship Mentor** – Mentoring early-career engineers through structured learning plans covering backend engineering, cloud, code reviews, and career guidance.

TECHNICAL SKILLS

- **Languages and Frameworks:** TypeScript, JavaScript, Python, SQL, Next.js, React, Node.js, Electron, Kotlin, Tailwind CSS
- **AI and Agents:** Model Context Protocol (MCP), LiteLLM, LangChain, LangFuse, OpenRouter, Ollama, RAG
- **Cloud and Infrastructure:** AWS, Azure, Terraform, Docker, PostgreSQL, PostHog, New Relic, WebSockets

EDUCATION

M.Tech Artificial Intelligence and Machine Learning CGPA: 8.77	BITS Pilani (WILP)	2025 – Present
B.Tech Information Technology CGPA: 9.34 · Anna University Rank Holder (Rank 6)	LICET Chennai	2019 – 2023

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate (AI-102)	Claude Certified Architect – Foundations
NVIDIA Certified Associate: AI Infrastructure & Operations	HashiCorp Certified: Terraform Associate
Microsoft Azure Fundamentals (AZ-900) · DP-900	Google Cloud Digital Leader

INDEPENDENT PROJECTS & CHALLENGES

- **RAG Visualizer** – Learning project to make retrieval-augmented generation flows easier to inspect and reason about.
- **AI News Generator** – Experiment exploring automated AI-news discovery, summarization, and delivery.
- **WIZ AI Security Challenge** and **WIZ Big IAM Challenge** – Hands-on cloud security and identity-access-management challenges.

RECOGNITION

- **Presidio Spot Award** for Generative AI platform contributions.
- **Winner, Creative Nomads** – frontend development competition.